VANIER
CÉGEP/COLLEGE

Faculty of Science and Technology
420-521-VA ADVANCED NETWORKS

Project Proposal

Project Name: **Building Firewall**

Team members

Daiyan Abrar Samin
Aaron Hossain

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Project Description

- We will explore how a firewall protects networks by controlling what packets go inside and outside the network.

- We will plan out the security policies and the rules that will enforce the policies.

- We will do this using Suricata (IDS) and test traffic that would trigger all the enforcements that we configure the firewall to stop.

- During our demonstration, we will show the firewall rules, logs and IDS alert.

Covered Subjects

Default deny: A default deny is an approach that blocks all traffic from accessing the network given specific rules and policies to protect the network.

Security policy: It is essentially a set of rules that define how the network is protected. It shows the expectations of how a firewall will block certain packets that are considered attacks.

Network zones: It allows the separation of certain parts of a network to maximize security. Different zones can have different enforcements and rules depending on the vulnerability of that certain area. For instance, DMZs are usually used for end users so only can access the part of the network the company intends and does not expose the entire structure of the network to prevent attacks.

Stateful filterting and connection tracking: Stateful filtering is a way of filtering packets that checks the entire connection traces instead of one single packet. It remembers the state of connections and makes decisions based on that information.

What a firewall can’t see: Basic firewalls can’t read encrypted application content or see any traffic that doesn’t go directly through it.

Intrusion detection: Intrusion detection is essentially detecting which packets are bad to the configured system and sends alerts to the developers. It's basically just monitoring network traffic to find out suspicious behaviour.

Softwares & Hardwares

- VirtualBox
- Ubuntu server (OS for the firewall)
- Nftables (for traffic rules)
- Suricata (intrusion detection)
- Curl (for connection testing)
- Tcpdump (for packet inspection)

Short Implementation Plan

1. Start and configure the VM.
2. Plan out the specific rules and policies.
3. Prepare the virtual network - here we will create the zones
4. Configure the firewall with the planned out policies.
5. Test the policies
6. Generate IDS alerts

References

Nftables in Ubuntu:
https://documentation.ubuntu.com/security/security-features/network/firewall/nftables/
Default deny: https://www.huntress.com/cybersecurity-101/topic/what-is-default-deny
Security policy: https://www.varonis.com/blog/what-is-a-security-policy
Network zones: https://www.lenovo.com/ca/en/glossary/what-is-zone/
Stateful filtering: https://www.sciencedirect.com/topics/computer-science/stateful-filtering
Firewall limits: https://www.fortinet.com/resources/cyberglossary/benefits-of-firewall
Intrusion detection:
https://www.geeksforgeeks.org/ethical-hacking/intrusion-detection-system-ids/
Suricata rules: https://docs.suricata.io/en/latest/rules/intro.html